

Research paper

Single-ended protection method of double-ended weak-infeed AC tie-line based on opposite-side voltage consistency analysis

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ABSTRACT

In order to solve the problem that the performance of traditional single-ended protection is degraded due to the common influence of the control strategy of the converters at both ends of the AC tie-line, this paper proposes a single-ended backup protection method suitable for the single-phase ground fault of the AC tie-line of the wind farm through VSC-HVDC system. For the single-ended protection on the wind farms side, firstly, the composite sequence network diagram of the AC tie-line single-phase ground fault is obtained according to the low voltage ride through (LVRT) strategy adopted by the wind farm and the flexible DC converter. Then, based on the obtained sequence network relationship, the voltage on the opposite side of the protected line is calculated, and compared with the voltage on the opposite side of the line calculated by the LVRT strategy of the flexible DC converter, and then the protection criterion for identifying internal and external faults is constructed. The results of the joint debugging of MATLAB and PSCAD/EMTDC simulation platform show that the voltage difference of internal fault is much smaller than that of external fault, which verifies the correctness of the protection criterion in this paper. As a new backup protection scheme for double-ended weak-infeed system, the proposed protection method can protect the entire line reliably and solve the dead zone problem existing in the traditional single-ended protection under the near-zone fault effectively, and has a great ability to withstand transition resistance, which can identify internal faults up to 300Ω.

1. Introduction

With the formulation of China's strategic goal of peaking carbon emissions before 2030 and achieving carbon neutrality before 2060, wind power is poised to become a crucial renewable energy generation method in the future (Shu et al., 2023; Zhou et al., 2014; Naidu et al., 2023). Because the VSC-HVDC transmission technology can avoid the resonance problem caused by long-distance AC transmission, and has lower transmission loss, higher reliability and higher transmission capacity than HVAC transmission (Chandio et al., 2023), it has become the preferred solution for power transmission of isolated island wind farms. VSC-HVDC also has most of the advantages of traditional HVDC, such as no stability problem of AC transmission, active and reactive power can be independently controlled by the control system (Chen et al., 2022), etc. At the same time, VSC-HVDC adopts full-controlled device IGBT and

high-frequency modulation technology, which is different from conventional HVDC using semi-controlled devices and low-frequency modulation technology, and has technical advantages (Fan et al., 2013; Yang et al., 2022). The VSC-HVDC transmission has been widely used in offshore wind power and onshore wind power in remote areas.

Different from the original wind farm directly connected to the power grid through the AC tie-line, in the grid-connected system of wind power through VSC-HVDC, both the wind farm and the flexible DC converter station are comprised of all power electronic equipment. In order to ensure the safety of the IGBT and other fully controlled devices, current limiting is the basic necessary link in the control system of the two. When a fault occurs in the AC tie-line between the wind farm and the flexible DC converter station, the double-ended power supply, due to the fault current supplied by it is generally limited to less than 1.2 times the rated current according to the tolerance capability of the power electronic devices (Yang et al., 2023), presents typical weak-infeed

Abbreviations: LVRT, The low voltage ride through strategy; VSC-HVDC, The voltage source converter based high voltage direct current transmission; DFIG, The doubly-fed induction generator; AC, Alternating current; DC, Direct current; HVAC, High voltage alternating current transmission; HVDC, High voltage direct current transmission; IGBT, Insulate-Gate Bipolar Transistor; MMC, Modular multilevel converter; PCC, Point of common coupling.

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Nomenclature

Symbols

$i_{rd+}^{+*}, i_{rq+}^{+*}$ (pu)	The reference values of positive sequence dq-axis current
$i_{rd-}^{+*}, i_{rq-}^{+*}$ (pu)	The reference values of negative sequence dq-axis current
u_{sd+}^{+} (pu)	The positive sequence component of the PCC voltage
u_{sd-}^{+}, u_{sq-}^{+} (pu)	The reference values of negative sequence dq-axis voltage
ω_1 (pu)	The rated angular frequency
k_d	The dynamic reactive power proportional coefficient of the wind farm
$i_{rq+}^{+ \max}$ (pu)	The maximum value of q-axis current command
i_{lim} (pu)	The maximum capacity of the converter
L_m (pu)	Excitation inductance
L_s (pu)	Inductance of DFIG stator
U_{PCC} (kV)	The positive sequence voltage at PCC point of the sending-end flexible DC converter
U_{acref} (kV)	The rated voltage of the AC side of the sending-end flexible DC converter
k	The voltage adjustment coefficient
I_{RMS} (kA)	The current measurement value
I_{lim} (kA)	The current limit
$U_{PMMC}^{+}, U_{PMMC}^{-}$ (kV)	The positive and negative sequence voltages at

PCC point of the flexible DC converter

U_W^{+}, U_W^{-} (kV)	The positive and negative sequence voltages at the protection installation
d	The percentage of the fault distance
I_W^{+}, I_W^{0} (kA)	The positive sequence current and zero sequence current collected by wind farm side bus W protection device
$I_M^{+}, I_M^{-}, I_M^{0}$ (kA)	The positive sequence, negative sequence and zero sequence current at PCC point of the flexible DC converter
$Z_{WM}^{+}, Z_{WM}^{-}, Z_{WM}^{0}$ ($10^{-2}\Omega/\text{km}$)	The positive sequence, negative sequence and zero sequence line equivalent impedance between bus W and M
$Z_{T3}^{+}, Z_{T3}^{-}, Z_{T3}^{0}$ (Ω)	The positive sequence, negative sequence and zero sequence reactance of transformer T3
Z_{T2}^{0} (Ω)	The zero sequence reactance of transformer T2
Z_{T31}^{+}, Z_{T31}^{0} (Ω)	The equivalent positive sequence, and zero sequence reactance from the transformer T3 fault grounding point to the star connection side grounding point.
I_{MMCj}^{+} (kA)	The three-phase current of the flexible DC converter ($j = a, b, c$)
k_{rel}	The reliability coefficient
U_n (kV)	The rated voltage

power supply characteristic. This characteristic is significantly different from the conventional synchronous power supply characteristic, resulting in a decline in the performance of traditional protection, and even causing protection misoperation or rejection (Lu et al., 2018; Song et al., 2023). Therefore, the fast and reliable operation of the wind farm-flexible DC converter AC tie-line protection of the double-ended weak-infeed power supply is of great significance for ensuring the efficient delivery of new energy. The single-ended protection, as an indispensable backup protection scheme in case of communication failure, needs to be studied in depth.

At present, some studies have been carried out on the single-ended protection of double-ended weak-infeed lines connected to the wind farm and flexible DC converter station. The influence of the wind farm and flexible DC converter control strategy on short-circuit current and existing protection is studied for the topology of permanent magnet wind farms transmitted by VSC-HVDC. However, the distinction between internal and external faults and the setting of zero-sequence current protection are not considered in (Zheng et al., 2021). In (Gao et al., 2023), a scheme is proposed to synergize the control strategy and protection of the flexible DC converter in order to solve the problem that the phase of fault current of flexible DC converter is uncontrollable, so as to avoid the phase difference of fault current between flexible DC converter and wind farms side exceeding the setting range. However, this scheme has the problem of conflict with the dynamic reactive power support requirements of new energy stations. In the study of (Zheng et al., 2023), an obvious difference between the negative sequence currents of faulted and non-faulted lines connected to the same bus is pointed out, and a double-ended weak-infeed system AC tie-line protection scheme is proposed based on the ratio of transient negative-sequence current accumulation. However, the control strategy of both wind farm and the flexible DC converter in this scheme adopts the negative sequence current suppression, and only constitutes the negative sequence loop with the help of the cable capacitance. For the single-phase grounding fault, it will be difficult to identify due to the small fault current. In (Li et al., 2023a), a distance protection scheme with the ability to withstand high transition resistance in the event of

single-phase ground fault on the island 110 kV AC line for the island system of new energy station-energy storage station is proposed. However, this scheme requires the control strategy support of the energy storage station.

From the above analysis, it can be seen that the single-ended protection of the double-ended weak-infeed system still exists shortcomings such as high dependence on the control strategy and great difficulty in setting, which restrict its practical promotion. Therefore, it is necessary to study a single-ended protection scheme which is independent of specific control strategy, easy to set and simple algorithm for the AC tie-line of the wind farm connected to the flexible DC system.

In this paper, a new single-ended backup protection scheme is proposed for the single-phase ground fault of the AC tie-line of the double-ended weak-infeed system. The main contributions are as follows:

1. Through the equivalent sequence network diagram of the wind power through the flexible DC transmission system when a single-phase ground fault occurs, the difference of the line opposite-side voltage between the internal and external faults in the AC tie-line is identified. According to this difference, a new single-ended protection method for double-ended weak-infeed AC tie-line based on the opposite-side voltage consistency analysis is proposed.
2. When the transition resistance is as high as 300 Ω , the calculation results are still accurate, indicating that the proposed protection method has a strong ability to resist transition resistance.
3. In this paper, the wind farm side low-through strategy first uses the positive sequence to meet the requirements of the inverter power supply for dynamic reactive power support, in line with the national standard. The voltage difference in the internal fault is much smaller than that in the external fault. The voltage difference between the internal and external faults is obvious and the setting is easy.
4. When a short-circuit fault occurs at the proximal end of the wind farm side protection device and the end of the line, the criterion proposed in this paper can act correctly, and the setting range covers the full length of the line, which has high reliability.

5. The protection method proposed in this paper is simple in principle and has universal applicability. When the control strategy of the MMC changes, as long as the control strategy is known, the voltage on the other side of the protected line can be calculated, and the protection method proposed in this paper is used to identify the internal and external faults. There is no need for specific control strategy support.

The rest of this paper is structured as follows. For the single-ended protection on the wind farms side, in Section 2, the LVRT control strategy adopted by the wind farm and the flexible DC converter is introduced. In Section 3, the opposite-side voltage of the protected line is calculated by using the composite sequence network relationship of the double-ended weak-infeed system, and compared with the voltage on the opposite side of the line calculated by the LVRT strategy of the flexible DC converter. In Section 4, the difference of the voltage on the opposite side of the protected line is analyzed when the internal and external faults occur. In Section 5, according to the voltage difference between internal and external faults, the fault identification criterion is constructed, and a new single-ended protection method based on the analysis of the voltage consistency of the opposite side is proposed. In Section 6, the effectiveness of the scheme is verified in PSCAD/EMTDC and MATLAB. Finally, the contributions and innovations of this paper are summarized in Section 7.

2. Fault ride-through control strategy of the wind farm and the flexible DC converter

The topology of the DFIG through VSC-HVDC system is shown in Fig. 1. In the sending end system, after the large-scale wind farm is collected through the 33 kV line, it is boosted to 230 kV through the main transformer and connected to the sending-end flexible DC converter through the AC tie-line. The sending-end converter then transmits the power to the receiving-end flexible DC converter through the DC line, which is subsequently connected to the receiving-end AC system.

The transient response time of the converter is usually within one to two cycles. Considering that its transient time is short, the converter is considered to reach steady state in this paper when calculating the voltage on the opposite side of the line (Wang and Kong, 2024). When the converter reaches steady state, the LVRT control strategy directly determines the composite sequence network diagram of the double-ended weak-infeed system. Therefore, this section introduces the LVRT control strategies of the doubly-fed induction generator (DFIG) and the sending-end flexible DC converter respectively.

2.1. Low voltage ride-through control strategy of the DFIG

The DFIG converter adopts the grid voltage-oriented vector control strategy. In normal operation, the rotor side converter adopts active and

reactive power control, and the grid side converter adopts DC bus voltage and reactive power control (Luo, 2021).

In the event of a fault, in order to ensure that the DFIG can still maintain operation under low voltage conditions, it is necessary to output a certain amount of reactive power to support the voltage recovery of the grid-connected point (Li et al., 2023b). At this time, the rotor side converter of the DFIG adopts the stator negative sequence current suppression control strategy to suppress the negative sequence current of the stator output to zero (Sun et al., 2023). Subsequently, according to the latest national standard GB/T 19963.1–2021, the reference value of the stator positive sequence q-axis current is preferentially modified. Then the reference value of the rotor positive sequence dq-axis current is determined according to the stator-rotor current relationship and the maximum allowable current of the rotor converter. The setting of the rotor dq-axis current reference value is given out as follows:

$$\begin{cases} i_{rd+}^{++} = \sqrt{i_{rlim}^2 - i_{rq+}^{++2} - i_{rd-}^{+2} - i_{rq-}^{+2}} \\ i_{rq+}^{++} = \min \left\{ -\frac{u_{sd+}^+}{\omega_1 L_m} - k_d (0.9 - u_{sd+}^+) \frac{L_s}{L_m}, i_{rq+}^{+max} \right\} \\ i_{rd-}^{+} = \frac{u_{sq-}}{\omega_1 L_m} \\ i_{rq-}^{+} = \frac{u_{sd-}}{\omega_1 L_m} \end{cases} \quad (1)$$

where i_{rd+}^{++} and i_{rq+}^{++} are the reference values of positive sequence dq-axis current; k_d is the dynamic reactive power proportional coefficient of the wind farm; u_{sd+}^+ is the positive sequence component of the PCC voltage; i_{rq+}^{+max} is the maximum value of q-axis current command; i_{rlim} is 1.2pu for the maximum capacity of the converter.

As can be observed from Eq. (1), according to the standard requirements, the dynamic reactive power proportional coefficient should be not less than 1.5 and not more than 3. Considering that the reactive power support of the DFIG is preferentially satisfied under low voltage, the maximum value of the q-axis current command value is set to the maximum allowable current of the converter 1.2pu.

When a fault occurs, the positive sequence target of the grid-side converter of the DFIG is still to maintain DC bus voltage stability as well as to control reactive power. The negative sequence adopts the current suppression control method, so that the negative sequence current output by the grid-side converter of the DFIG is zero. At this time, the stator-side and grid-side converters have no negative sequence current output, which results in no negative sequence of the total output current of the DFIG.

2.2. LVRT control strategy of the sending-end flexible DC converter

Since the DFIG requires a uniform voltage and frequency reference

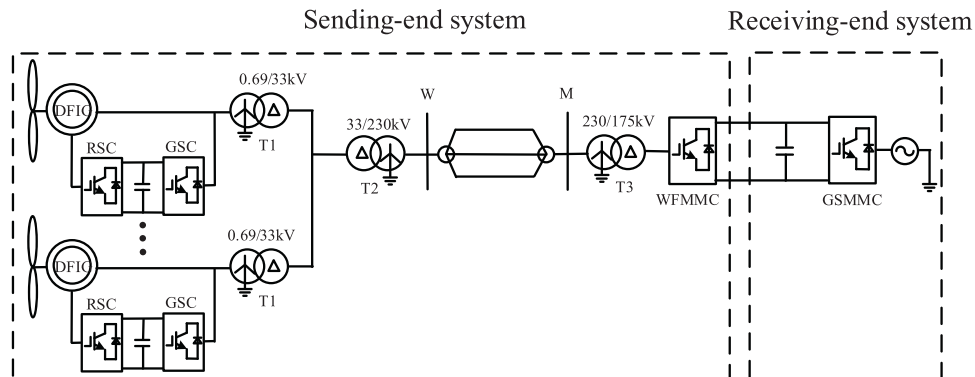


Fig. 1. Topology diagram of the DFIG through VSC-HVDC system.

from the external system to complete the electrical energy collection and transmission, the sending-end flexible DC converter is directly connected to the DFIG through the AC tie-line, which usually adopts V/f control (Peng et al., 2023). When the AC tie-line faults, in order to prevent overcurrent in the bridge arm of the sending-end flexible DC converter, a positive sequence low voltage current limiting control strategy is adopted (Zheng et al., 2021). The positive sequence voltage reference value is reduced by constraining the line phase currents, thus limiting the bridge arm current after the fault. The voltage and current relationship of the positive sequence low voltage current limiting control strategy is given out as follows:

$$U_{PCC}^+ = U_{acref} - k(I_{RMS} - I_{lim}) \quad (2)$$

where U_{PCC}^+ is the positive sequence voltage at PCC point of the sending-end flexible DC converter; U_{acref} is the rated voltage of the AC side of the sending-end flexible DC converter; k is the voltage adjustment coefficient; I_{RMS} is the current measurement value; I_{lim} is the current limit.

It can be seen from Eq. (2) that after the single-phase ground fault occurs, the short-circuit current of the flexible DC converter increases transiently, resulting in a decrease in its positive sequence voltage reference value. With the decrease of the positive sequence voltage reference value, the output current of the control system reduces accordingly, thus reducing the short-circuit current, and there exists a dynamic balance value of the voltage and current, so that the short-circuit current does not exceed the maximum current allowed by the flexible DC converter bridge arm under this value. At the same time, the positive sequence voltage reference value is greater than zero, and the system reaches the fault steady state. By limiting the positive sequence voltage reference value of the flexible DC converter, the short-circuit current provided by the flexible DC converter can be limited, and then the LVRT of the sending-end flexible DC converter can be realized.

In this section, the control system of the DFIG converter and the sending-end flexible DC converter is summarized as shown in Fig. 2.

3. Calculation of the voltage on the line opposite-side of the single-phase ground fault in the area

Taking the composite sequence network diagram and the LVRT strategy of the flexible DC converter under the single-phase ground fault

in the area as the research object, the opposite-side voltage is calculated, and the expression between the percentage of fault distance and the positive sequence voltage at the outlet of the flexible DC converter is derived.

3.1. Calculation of the opposite-side voltage based on composite sequential network relationship

It is assumed that the A-phase ground fault occurs on the 230 kV AC tie-line of the DFIG connected to the flexible DC system. For the single-ended protection close to the wind farm side, the ground fault occurs in the area. The voltage and current quantities at the installation of the protection device can be measured, so the bus voltage and line current on the wind farm side can be regarded as known quantities, and the whole of its back side can be regarded as a voltage source with a known output current (Edrah et al., 2015).

According to the LVRT strategy introduced in Section 2, the sending-end flexible converter can be equivalent to a current-controlled voltage source, and the equivalent three-phase circuit diagram is shown in Fig. 3.

where, $\dot{E}_{Wa}$, $\dot{E}_{Wb}$ and $\dot{E}_{Wc}$ are the bus voltage at the protection device on the wind farm side; $\dot{I}_{Wa}$, $\dot{I}_{Wb}$ and $\dot{I}_{Wc}$ are the line current at the protection device; $\dot{I}_{MMCa}$, $\dot{I}_{MMCb}$ and $\dot{I}_{MMCc}$ are the phase current of each phase

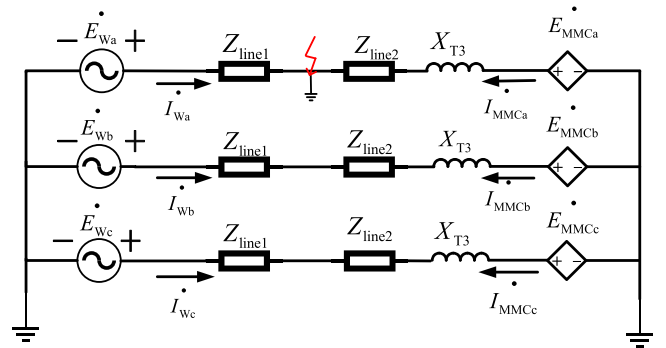


Fig. 3. Equivalent three-phase circuit diagram for internal fault.

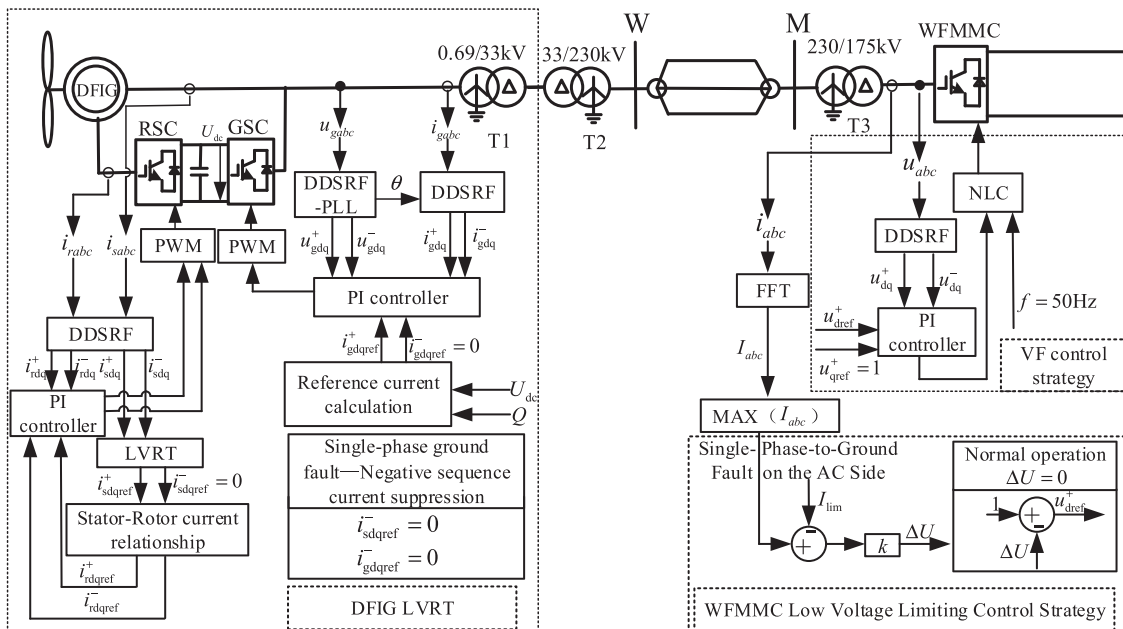


Fig. 2. Control system structure diagram of the DFIG converter and flexible DC converter.

for the flexible DC converter; Z_{line1} is the line impedance from the wind farm side bus W to the fault point and Z_{line2} is the line impedance from the fault point to the flexible DC converter side bus M.

According to the symmetrical component method, the three-phase circuit diagram of the fault steady state can be transformed into a composite sequence network diagram. Considering that the wind farm side adopts the negative sequence current suppression strategy, the negative sequence current on the wind farm side is zero in the sequence network diagram, which is shown as an open circuit form (Geng et al., 2013). The flexible DC converter adopts a positive sequence low voltage current limiting control strategy, which is equivalent to a current-controlled voltage source in the positive-sequence network. Since there is no direct control of the negative sequence voltage of the flexible DC converter, the flexible DC converter is equivalent to a voltage source in the negative sequence network, and its voltage magnitude and phase are unknown. The composite sequence network diagram of the double-ended weak-infeed system when the A-phase ground fault occurs in the area is shown in Fig. 4.

where, $\dot{U}_W^+$ and $\dot{U}_W^-$ are bus positive sequence and negative sequence voltages at the protection installation; $\dot{U}_{PMMC}^+$ and $\dot{U}_{PMMC}^-$ are positive and negative sequence voltages at PCC point of the flexible DC converter; Z_{WM}^+ , Z_{WM}^- and Z_{WM}^0 are the positive sequence, negative sequence and zero sequence line equivalent impedance between bus W and M; Z_{T3}^+ , Z_{T3}^- and Z_{T3}^0 are the positive sequence, negative sequence and zero sequence reactance of transformer T3; d is the ratio of the fault distance from the fault point to the bus W protection installation to the length of the bus WM line, that is, the percentage of the fault distance.

According to Kirchhoff's voltage law, the positive sequence, negative sequence and zero sequence circuit networks in the sequence network diagram of Fig. 4 can be listed with their corresponding voltage equations as shown in Eqs. (3–5) respectively.

$$\dot{U}_W^+ - \dot{I}_W^+ dZ_{WM}^+ - \dot{U}_{PMMC}^+ - \dot{I}_M^+ [Z_{T3}^+ + (1-d)Z_{WM}^+] = 0 \quad (3)$$

$$\dot{U}_W^- - \dot{U}_{PMMC}^- - \dot{I}_M^- [Z_{T3}^- + (1-d)Z_{WM}^-] = 0 \quad (4)$$

$$(Z_{T2}^0 + dZ_{WM}^0)\dot{I}_W^0 - [Z_{T3}^0 + (1-d)Z_{WM}^0]\dot{I}_M^0 = 0 \quad (5)$$

Because the sequence network diagram of single-phase ground fault is three-sequence network in series, the relationship between positive sequence, negative sequence and zero sequence current can be obtained

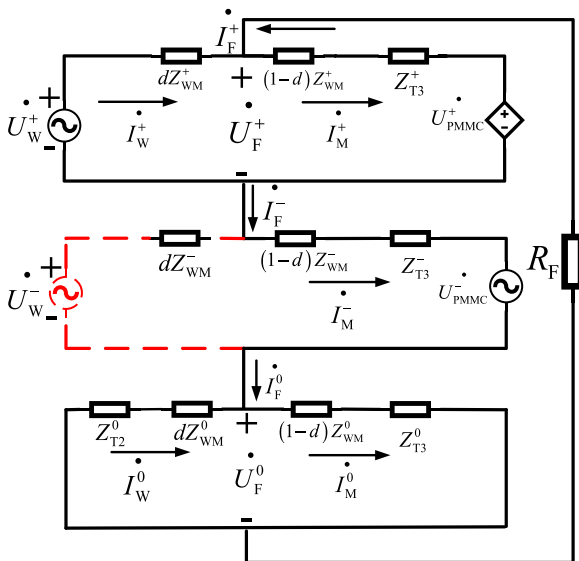


Fig. 4. Composite sequence network diagram for internal fault.

as follows:

$$\dot{I}_M^+ - \dot{I}_W^+ = \dot{I}_M^- \quad (6)$$

$$\dot{I}_W^+ + \dot{I}_M^0 = \dot{I}_M^- \quad (7)$$

The relationship between the zero-sequence current of the flexible DC converter and the positive sequence and zero sequence of the wind farm side and the positive sequence current of the flexible DC converter can be obtained by associating Eq. (6) and Eq. (7), and the relationship between the positive sequence current of the flexible DC converter and the positive sequence and zero sequence current of the wind farm side can be obtained by substituting this relationship into Eq. (5), as shown in Eq. (8).

$$\dot{I}_M^+ = \dot{I}_W^+ + \frac{(Z_{T2}^0 + Z_{T3}^0 + Z_{WM}^0)}{Z_{T3}^0 + (1-d)Z_{WM}^0} \dot{I}_W^0 \quad (8)$$

The relationship between the positive sequence voltage at the PCC point of the flexible DC converter and the percentage of fault distance can be obtained by substituting the Eq. (8) into the Eq. (3), as shown in Eq. (9).

$$\dot{U}_{PMMC}^+ = \dot{U}_W^+ - \dot{I}_W^+ Z^+ - \dot{I}_W^0 Z^0 \frac{Z_{T3}^+ + (1-d)Z_{WM}^+}{Z_{T3}^0 + (1-d)Z_{WM}^0} \quad (9)$$

where

$$\left. \begin{aligned} Z^+ &= Z_{T3}^+ + Z_{WM}^+ \\ Z^0 &= Z_{T2}^0 + Z_{T3}^0 + Z_{WM}^0 \end{aligned} \right\} \quad (10)$$

For Eqs. (3–7), $\dot{U}_W^+$, $\dot{U}_W^-$, $\dot{I}_W^+$ and $\dot{I}_W^0$ are known quantities of positive sequence voltage, negative sequence voltage, positive sequence current and zero sequence current collected by wind farm side bus W protection device. $\dot{U}_{PMMC}^+$ and $\dot{U}_{PMMC}^-$ are positive sequence and negative sequence voltage at PCC point of the flexible DC converter. $\dot{I}_M^+$, $\dot{I}_M^-$ and $\dot{I}_M^0$ are positive sequence, negative sequence and zero sequence current at PCC point of the flexible DC converter. d is the percentage of the fault distance. These are all unknown quantities. Therefore, the expression between the percentage of fault distance and the positive sequence voltage at the outlet of the flexible DC converter can be obtained after derivation as shown in Eq. (9).

3.2. Calculation of the opposite-side voltage based on the LVRT strategy of the flexible DC converter

In order to distinguish it from the output voltage of the flexible DC converter in the composite sequence network diagram, the calculation of the opposite-side voltage based on the LVRT strategy of the flexible DC converter in this section adopts $\dot{U}_{PMMC}^+$.

Through the derivation of the sequence network diagram circuit in Subsection 3.1, it can be seen that only the expression of the relationship between the percentage of fault distance and the positive sequence voltage at the outlet of the flexible DC converter is obviously insufficient if the protection criterion is to be constituted. Therefore, based on the positive sequence low voltage current limiting control strategy of the flexible DC converter introduced in Section 2, the relationship between the percentage of fault distance and the positive sequence voltage at the outlet of the flexible DC converter is derived.

From Section 2, it can be seen that according to the detected maximum phase current, the positive sequence d-axis voltage reference value of the flexible DC converter can be obtained by using the Eq. (2), in which its positive sequence q-axis voltage reference value is consistent with the normal operation, which is zero. Considering the relationship to be derived, the LVRT strategy can be written as the relationship

between the phase current of the bridge arm and the positive sequence voltage, which is shown in Eq. (11).

$$|\dot{U}_{PMMC}^+| = U_{acref} - k \left(\left| \max \{ I_{MMCa}, I_{MMCb}, I_{MMCc} \} \right| - I_{lim} \right) \quad (11)$$

where U_{acref} is the voltage value in normal operation, which is generally set as the rated voltage; k is the voltage adjustment coefficient; I_{MMCa} , I_{MMCb} and I_{MMCc} are the three-phase current of the flexible DC converter; I_{lim} is the current limit value, and generally can be taken 1.2 times the rated current.

According to the transformation matrix of the symmetrical component method, the phase current of the flexible DC converter can be transformed into the relationship of the three-sequence current, and its expression is shown in Eq. (12).

$$\begin{cases} I_{MMCa} = \dot{I}_M^+ + \dot{I}_M^- \\ I_{MMCb} = \alpha^2 \dot{I}_M^+ + \alpha \dot{I}_M^- \\ I_{MMCc} = \alpha \dot{I}_M^+ + \alpha^2 \dot{I}_M^- \end{cases} \quad (12)$$

The relationship between the positive sequence voltage at the PCC point of the flexible DC converter and its positive and negative sequence current can be obtained by associating Eqs. (11) and (12), as shown in Eq. (13).

$$|\dot{U}_{PMMC}^+| = U_{acref} - k \left(\left| \dot{I}_{MMC}^+ \right| - I_{lim} \right) \quad (13)$$

where

$$\dot{I}_{MMC} = \max \left\{ \left(\dot{I}_M^+ + \dot{I}_M^- \right), \left(\alpha^2 \dot{I}_M^+ + \alpha \dot{I}_M^- \right), \left(\alpha \dot{I}_M^+ + \alpha^2 \dot{I}_M^- \right) \right\} \quad (14)$$

By associating Eqs. (6) and (8) in Subsection 3.1, the relationship between the negative sequence current of the flexible DC converter and the zero sequence current on the wind farm side can be obtained, as shown in Eq. (15).

$$\dot{I}_M^- = \frac{(Z_{T2}^0 + Z_{T3}^0 + Z_{WM}^0) \dot{I}_W^0}{Z_{T3}^0 + (1-d)Z_{WM}^0} \quad (15)$$

By substituting Eqs. (8), (14) and (15) into Eq. (13), the relationship equation between positive sequence voltage at PCC point and the percentage of fault distance under the LVRT strategy of the flexible DC converter can be obtained as shown in Eq. (16).

$$|\dot{U}_{PMMC}^+| = U_{acref} - k \left(\left| \max \left\{ \left(\dot{I}_W^+ + 2f(d)\dot{I}_W^0 \right), \left(\alpha^2 \dot{I}_W^+ - f(d)\dot{I}_W^0 \right), \left(\alpha \dot{I}_W^+ - f(d)\dot{I}_W^0 \right) \right\} \right| - I_{lim} \right) \quad (16)$$

where

$$f(d) = \frac{(Z_{T2}^0 + Z_{T3}^0 + Z_{WM}^0)}{Z_{T3}^0 + (1-d)Z_{WM}^0} \quad (17)$$

According to Eq. (16), the expression between the outlet positive sequence voltage under the positive sequence low voltage current limiting control strategy of the flexible DC converter and the percentage of fault distance can be known.

4. Analysis of the voltage on the line opposite-side of the single-phase ground fault outside the area

According to the topological structure diagram of the DFIG through

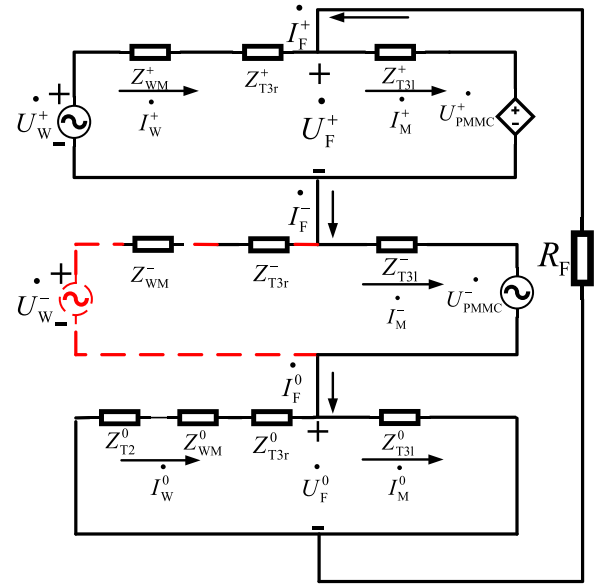


Fig. 5. Composite sequence network diagram for external fault.

VSC-HVDC system in Fig. 1, it can be seen that for the protection device at the bus W, the single-phase ground fault between the 230 kV AC tie-line is an internal fault. The fault at the bus M to the PCC point of the flexible DC converter outlet is an external fault, which mainly includes the internal ground fault of the transformer T3 high-voltage side.

When the fault occurs within the transformer T3, considering the internal grounding of the high-voltage side of the transformer, if the voltage on the opposite side is still calculated according to the composite sequence network diagram, the equivalent reactance of the sequence network diagram transformer T3 will be divided into two parts. The equivalent composite sequence network diagram is shown in Fig. 5.

where, Z_{T3r}^+ , Z_{T3r}^- and Z_{T3r}^0 are the equivalent positive sequence, negative sequence and zero sequence reactance from AC tie-line to the grounding point; Z_{T3l}^+ , Z_{T3l}^- and Z_{T3l}^0 are the equivalent positive sequence, negative sequence and zero sequence reactance from the transformer T3 fault grounding point to the star connection side grounding point.

According to the composite sequence network diagram of Fig. 5, the Kirchhoff voltage equation and current equation are listed, and the calculation idea of the opposite side voltage in Subsection 3.1 is still adopted. When the fault occurs outside the area, the relationship be-

tween the outlet voltage of the flexible DC converter and the fault distance can be written as shown in Eq. (18).

$$\dot{U}_{PMMC}^+ = \dot{U}_W^+ - \dot{I}_W^+ Z^+ - \dot{I}_W^0 Z^0 \frac{Z_{T3l}^+}{Z_{T3l}^0} \quad (18)$$

where

$$\begin{cases} Z^+ = Z_{T3}^+ + Z_{WM}^+ \\ Z^0 = Z_{T2}^0 + Z_{T3}^0 + Z_{WM}^0 \end{cases} \quad (19)$$

When the external fault occurs, if the opposite-side voltage calculated based on the composite sequence network relationship is consistent with the opposite-side voltage calculated based on the LVRT strategy of the flexible DC converter, the voltage calculation should be

carried out by using the Eq. (16) and Eq. (18). The difference between Eq. (9) and Eq. (18) can be found by comparing two equations. In Eq. (9), the coefficient related to the zero-sequence current is the ratio of the sum of the positive sequence impedance of the line and the positive sequence reactance of the transformer T3 to the sum of the zero-sequence impedance. While in Eq. (18), the coefficient related to the zero-sequence current is the ratio of the equivalent positive sequence reactance and the equivalent zero-sequence reactance from the fault grounding point of the transformer T3 to the grounding point of the star-side. There is a significant difference between these two equations. Therefore, for the external fault, the calculation of the opposite side voltage is still carried out by the Eq. (9). No matter how the fault distance changes from 0 to 1, the calculation results will be significantly different from those of Eq. (18).

5. Single-ended protection process based on the opposite-side voltage consistency

When a single-phase ground fault occurs, the fault distance is traversed from 0 to 1. For the internal fault, there must exist a fault distance between 0 and 1 to ensure that the opposite side voltage calculated by the composite sequence network relationship is equal to the opposite side voltage calculated by the LVRT strategy of the flexible DC converter. For the external fault, no matter how the fault distance changes between 0 and 1, the two voltage values are not equal. Therefore, the internal and external fault can be distinguished according to the voltage difference between two voltage values. The protection criterion is shown in Eq. (20).

$$\min \left| U_{PMSC}^+(d) - U_{PMSC}^-(d) \right| \leq k_{rel} U_n \quad (20)$$

where k_{rel} is the reliability coefficient; U_n is rated voltage.

The action flow of the single-ended protection based on opposite-side voltage consistency analysis in this paper is as follows:

Step 1, Data processing: The voltage and current measured by the protection device are transformed by full-wave Fourier transform and symmetrical component method to obtain the amplitude and phase of the three-sequence voltage and current.

Step 2, Equation calculation: Substituting the processed electrical quantities and line impedance parameters into Eq. (9) and Eq. (16), the expression of the voltage on the opposite side of the AC tie-line and the fault distance is obtained.

Step 3, Protection judgment: The fault distance is changed from 0 to 1, and the minimum value of the absolute value of the voltage difference in this range is calculated by Eq. (9) and Eq. (16). The minimum value is compared with the setting value. When it is less than the setting value, it is judged as an internal fault and the protection correctly acts. When it is greater than the setting value, it is judged to be an external fault, and the protection correctly does not act.

The flow chart of the protection action of the double-ended weak-infeed system is shown in Fig. 6.

6. Simulation verification

In order to verify the correctness of the protection criterion proposed in this paper, a simulation model of the double-ended weak-infeed system is established in PSCAD/EMTDC, as shown in Fig. 1. In this model, the rated capacity of a single DFIG is 5 MW, with a total of 20 units, and the terminal voltage of the wind turbine is 0.69 kV. The wind farm is boosted to 230 kV by the main transformer, connected to the sending-end flexible DC converter through the 100 km AC tie-line, and sent to the receiving-end flexible DC converter through the DC cable with a rated voltage of 500 kV to be integrated into the AC power grid. The 230 kV AC tie-line adopts the frequency-dependent model. The positive sequence impedance parameter of the overhead line under

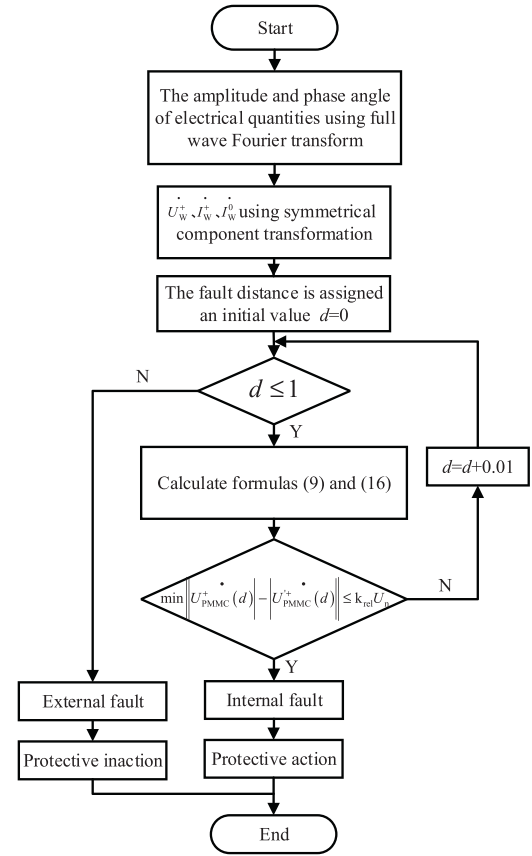


Fig. 6. Double-ended weak-infeed system protection action flow chart.

power frequency is $(0.03513 + j0.4015) \Omega/\text{km}$, and the zero-sequence impedance parameter is $(0.2827 + j1.187) \Omega/\text{km}$. Both the DFIG and the sending-end flexible DC converter use the LVRT control strategy as shown in Fig. 2.

For the internal fault, this simulation sets single-phase ground faults at 0 km, 20 km, 40 km, 60 km, 80 km, and 100 km of the 230 kV overhead line from the bus W protection side. The transition resistance is set to be 0.1Ω, 150Ω, and 300Ω at different fault locations. For the external fault, a ground fault is set inside the high-voltage side of the outlet transformer of the sending-end flexible DC converter. The fault location of the internal and external fault is shown in Fig. 7.

When a single-phase ground fault occurs at 5.5 s, the voltage and current collected by the wind farm protection device are saved to MATLAB, and the calculation results of the side voltage are verified according to the single-ended protection action flow in Section 5. The reliability coefficient is set to 0.05, and the setting value is 11.5 kV. The fault steady-state data after 1.5 s of fault occurrence is selected for simulation verification. The sampling frequency of voltage and current is 5 kHz. The results of the voltage difference are shown in Figs. 8 and 10.

6.1. Validation of the calculation results of the voltage on the opposite side in case of internal fault

For the internal fault, that is, single-phase ground faults occur between line WMs. Under the same fault point, by changing the magnitude of the grounding transition resistance, the absolute value of the calculated voltage difference on the opposite side is shown in Fig. 8. In order to see the result curve of in Figs. 8(d), 8(e), 8(f) more clearly, its ordinate is enlarged to 0–14.

From Fig. 8, it can be seen that when a metallic single-phase ground fault occurs in the area, a near-zone fault occurs at a distance of 0 km

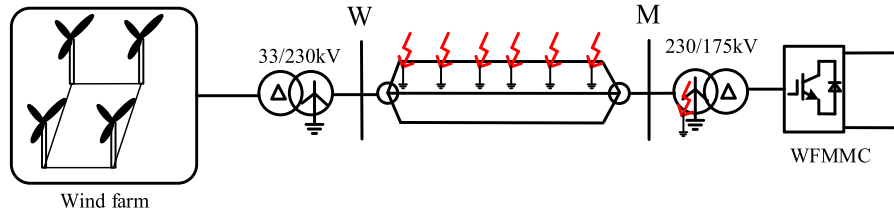


Fig. 7. Double-ended weak-infeed system simulation fault location diagram.

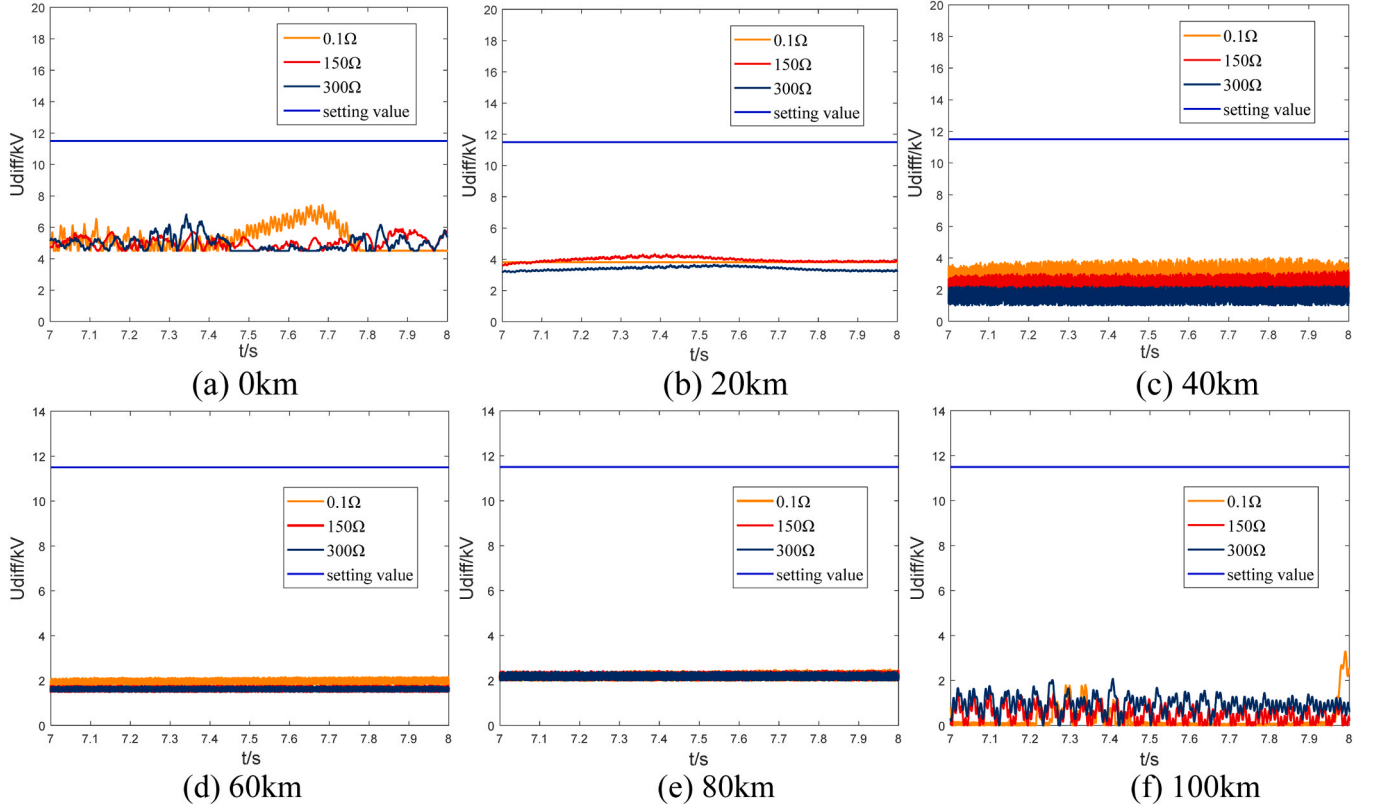


Fig. 8. Double-ended weak-infeed system simulation fault location diagram.

from the protection device, and the absolute value of the voltage difference on the opposite side fluctuates around 5 kV, which is less than the preset setting value of 11.5 kV. At 20 km and 40 km away from the protection device, the absolute value of the voltage difference on the opposite side fluctuates around 4 kV, which is also less than the preset setting value. As the fault location gradually moves away from the protection device, the absolute value of the voltage difference on the opposite side is basically stable at 2 kV at 60 km and 80 km, again less than the setting value. When the fault occurs at 100 km, that is, the full length of the line, the absolute value of the voltage difference on the opposite side fluctuates around 1 kV, which is much smaller than the setting value. In summary, whether it is a near-zone fault or a line-end fault, the absolute value of the voltage difference on the opposite side is less than the preset setting value. This shows that the criterion proposed in this paper solves the dead zone problem of short-circuit fault at the protection device. As the fault location gradually moves away from the protection device, the absolute value of the opposite side voltage difference also shows a gradual decrease trend, which avoids the problem of rejection when the fault occurs at the protection boundary.

Since it is not sufficient to only use the difference of the opposite side voltage as the fault information, it is necessary to consider whether the opposite side voltage obtained based on the composite sequence network relationship and the opposite side voltage obtained based on

the LVRT strategy of the flexible DC converter after the fault is consistent with the actual measurement value of the opposite side voltage to verify the accuracy of the Eq. (9) and Eq. (14) derived in this paper. In the simulation model, a single-phase metallic ground fault is set in the area, and the fault location is still set at an equal interval of 20 km from 0 to 100 km. By giving the fault distance in Eq. (9) and Eq. (14), the positive sequence voltage, positive sequence current and zero sequence current collected at the protection device are used to calculate the opposite side voltage at the fault location. The comparison between the estimated value of the opposite side voltage and the actual voltage measurement value is shown in Fig. 9.

In Fig. 9, it can be seen that the opposite side voltage based on the composite sequence network relationship in the protection criterion is in good agreement with the actual measured value. The opposite-side voltage obtained based on the MMC LVRT strategy is slightly larger than the opposite-side voltage obtained based on the composite sequence network relationship and the actual measured value, but the difference is also within 6 kV, which is less than the set value of 11.5 kV. This shows that the opposite side voltage calculated by the Eq. (9) and Eq. (14) is consistent with the actual opposite side voltage, and the difference of the opposite side voltage obtained from the Eq. (9) and Eq. (14) can be used as the protection criterion for protection judgment.

Since Eq. (9) and Eq. (16) in Section 3 are independent of the

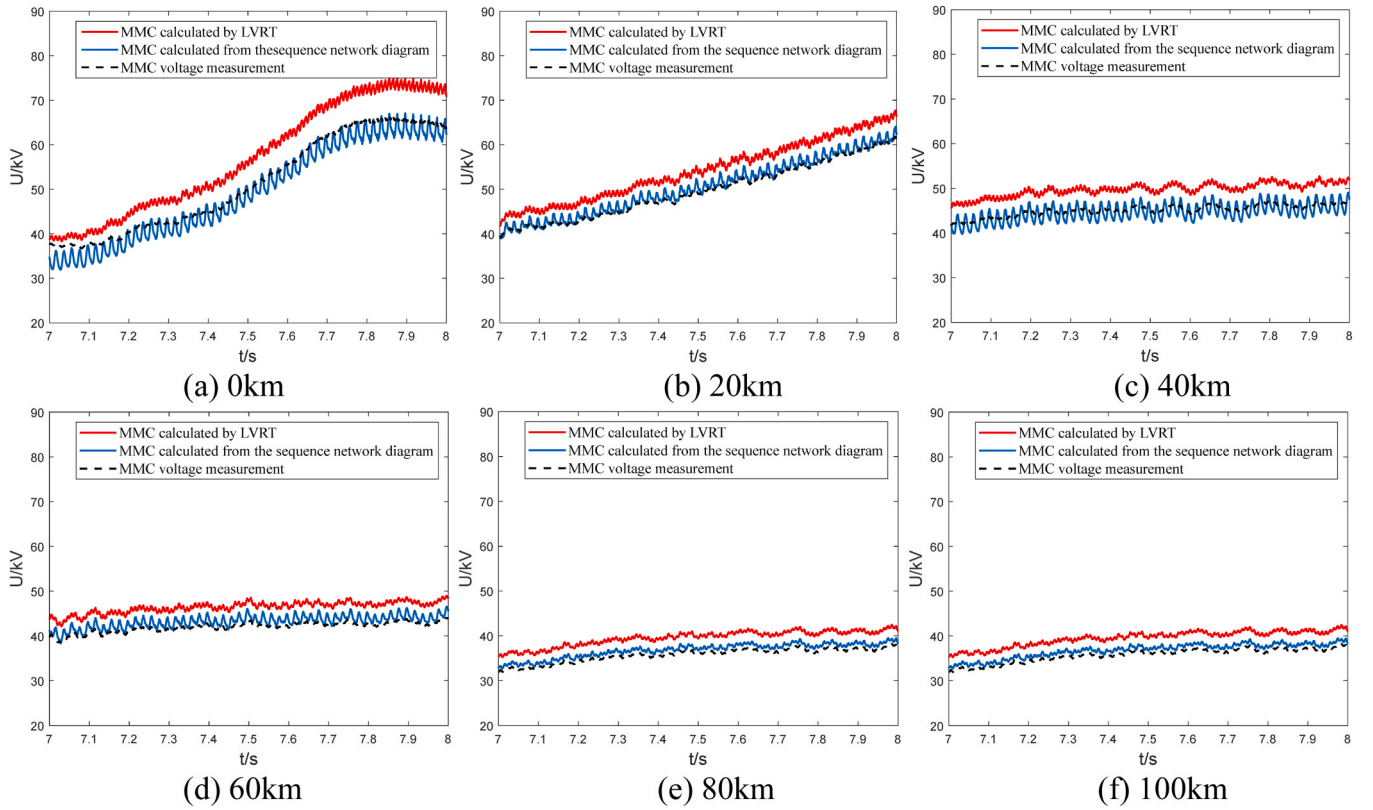


Fig. 9. Verification of MMC fault voltage estimation value and actual voltage measurement value.

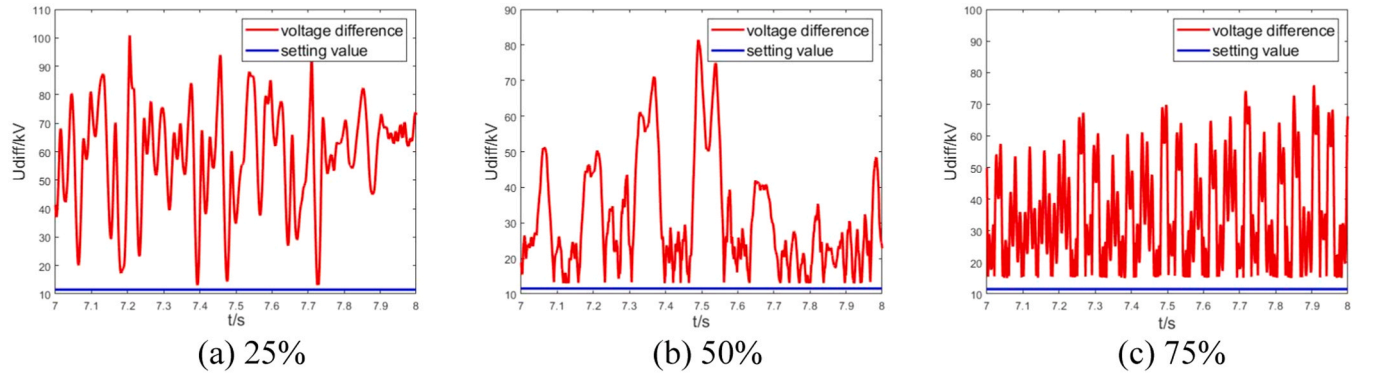


Fig. 10. Calculation of voltage on the opposite side in case of the external fault.

transition resistance, it shows that when the fault location is unchanged, whether it is a metallic grounding fault or a nonmetallic grounding fault, the fault distance of the minimum voltage difference on the opposite side is approximately equal. This can also be verified by Fig. 8. At the same fault point, when the transition resistance is 0.1Ω , 150Ω and 300Ω respectively, the calculation results of the opposite side voltage are equal, that is, the single-ended protection criterion based on the consistency of the opposite side voltage proposed in this paper is not affected by the transition resistance.

In order to more clearly and intuitively verify the accuracy of the protection criterion under different transition resistance during the internal fault, the fault distance that minimizes the voltage difference on the opposite side can be obtained, and the absolute error and relative error are defined as shown in Eq. (21).

$$\begin{cases} e_a = x_j - x_s \\ e_r = \frac{e_a}{l} \times 100\% \end{cases} \quad (21)$$

where x_j is the result of fault distance calculation; x_s is the actual fault distance; l is the total length of the transmission line.

The calculation results of fault distance at different fault locations are shown in Table 1.

In Table 1, it can be seen that the fault distance can be accurately calculated by using Eq. (9) and Eq. (16). However, when the fault occurs at 0 km away from the protection device, the error of the fault distance calculation result is large, reaching 20 %. This is due to the fact that the fault location is close to the wind farm side, which is limited by the current and voltage resistance of the wind turbine converter. The control system has limited adjustment ability and cannot track the original command value, resulting in the change of the circuit relationship of the

Table 1
Fault distance error table of 100 km overhead transmission line.

transition resistance	fault distance/ km	calculation result/km	absolute error/km	relative error
0.1Ω	0	18.82	18.82	18.82 %
	20	32.18	12.18	12.18 %
	40	45.32	5.32	5.32 %
	60	62.49	2.49	2.49 %
	80	80.17	0.17	0.17 %
	100	100	0	0
150Ω	0	9.69	9.69	9.69 %
	20	29.14	9.14	9.14 %
	40	42.19	2.19	2.19 %
	60	61.33	1.33	1.33 %
	80	79.72	-0.28	-0.28 %
	100	100	0	0
300Ω	0	7.81	7.81	7.81 %
	20	26.96	6.96	6.96 %
	40	41.77	1.77	1.77 %
	60	61.25	1.25	1.25 %
	80	80.32	0.32	0.32 %
	100	100	0	0

composite sequence network diagram, which makes the fault distance error calculated by the original composite sequence network diagram large. However, it can be seen from Fig. 8 that when a near-zone fault occurs, the absolute value of the voltage difference on the opposite side is still less than the setting value. This indicates that although the error of the fault distance calculation result is large when the near-zone fault occurs, it does not affect the judgment result of the protection criterion.

6.2. Validation of the calculation results of the voltage on the opposite side in case of external fault

In actual simulation equivalent model, the three-phase double-winding transformer T3 is composed of three single-phase double-winding transformers. For the external fault, when a ground fault occurs on the high-voltage side of the three-phase double-winding transformer T3, the fault part is regarded as the third winding, which is equivalent to changing the double-winding transformer of the fault phase into a three-winding transformer. By changing the size of the leakage reactance of the third winding, ground faults at different locations are realized. The fault points are set at 25 %, 50 % and 75 % of the high voltage side of the transformer T3. The calculation results of the voltage on the opposite side are shown in Fig. 10.

In Fig. 10, it can be seen that when the external fault occurs, the calculated result of the opposite side voltage fluctuates greatly, and the minimum value of the calculated result of the opposite side voltage cannot be clearly known. In order to make a numerical comparison between the minimum voltage and the setting value, and verify whether

the criterion proposed in this paper can correctly identify the external fault, the result diagram of Fig. 10 is partially enlarged, and the enlarged result diagram is shown in Fig. 11. In these figures, red is still the opposite side voltage difference, and blue is the setting value.

In Fig. 11, it can be seen that the minimum value of the calculation results when the fault occurs at 25 % and 50 % is about 12 kV, and the minimum value of the calculation result when the fault occurs at 75 % is about 15 kV. The minimum value of voltage difference is greater than the setting value of 11.5 kV, which indicates that the criterion proposed in this paper can reliably not operate when an external fault occurs. Although the minimum value is greater than the setting value, in order to further improve the reliability of the protection, the maximum and minimum values of the calculation results are selected and averaged as the absolute value of the opposite side voltage difference in the criterion of external fault. The final average result is shown in Fig. 12.

In Fig. 12, it can be seen that when the fault occurs at 25 % of the high voltage side of the transformer T3, the average result is 68 kV, which is much larger than the setting value of 11.5 kV. When the fault occurs at 50 % and 75 % of the high voltage side of the transformer T3, the average result is about 48 kV, which is also much larger than the setting value of 11.5 kV. It shows that using the proposed criterion, both the calculation result before averaging and the result after averaging are larger than the setting value. The criterion can correctly recognize the external fault.

6.3. Comparison with existing protection method

The scenario studied in (Li et al., 2023a) is an island system of a new energy station-energy storage station, and both ends are inverter power sources, which is the same as the scenario studied in this paper. The research method of (Li et al., 2023a) is to use the negative sequence open-circuit characteristics of the new energy station, use the local negative sequence current to calculate the fault point current, and propose a distance protection scheme, which is suitable for the double-ended inverter power supply line. However, (Li et al., 2023a) did not conduct simulation verification for the near-zone fault.

The protection method proposed in this paper is compared with that proposed in (Li et al., 2023a). When a ground fault occurs at 0 km away from the protection device in the positive direction of the wind farm side protection device, the transition resistance is 0.1Ω, 150Ω and 300Ω, respectively. The calculation results of the two protection methods are shown in Table 2.

In Table 2, it can be seen that when a single-phase grounding fault occurs at the protection device, whether it is a metallic grounding fault or a high-resistance fault, the protection proposed in (Li et al., 2023a) cannot operate correctly, and the calculated fault distance is a negative value. The error between the calculated fault distance and the actual fault distance is very large, which exceeds the protection range of the

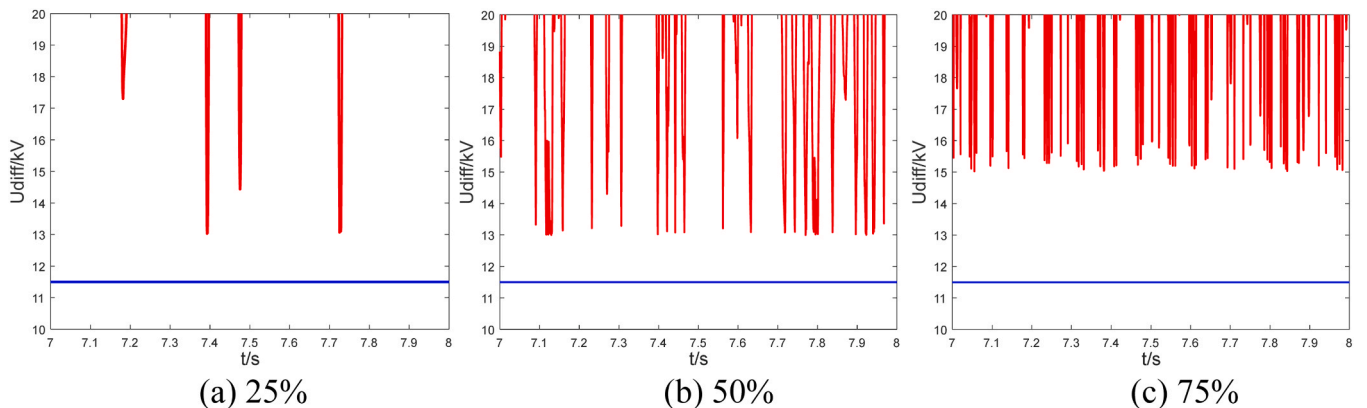


Fig. 11. Amplification diagram of the calculation results of the opposite side voltage in case of the external fault.

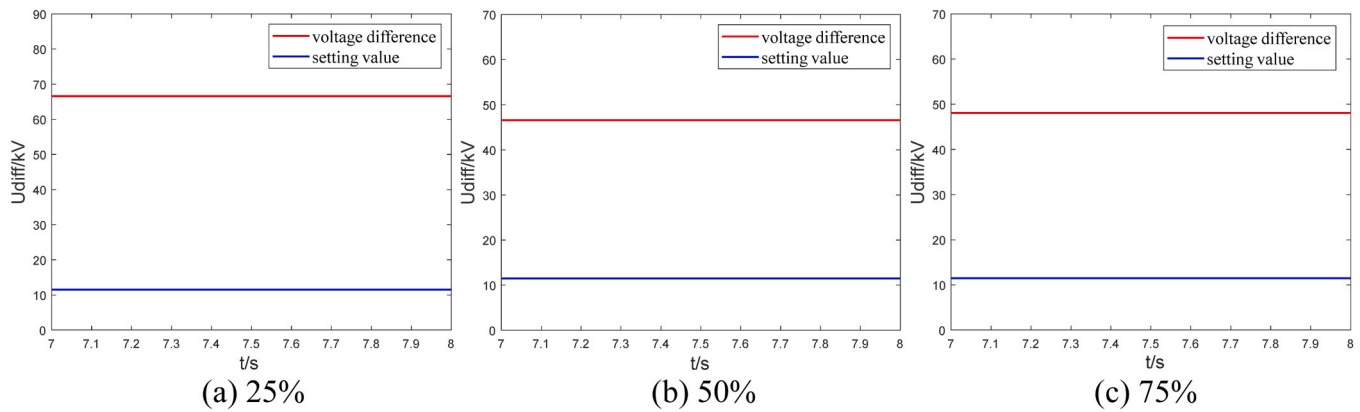


Fig. 12. Results after averaging in case of the external fault.

Table 2

Comparison between the proposed protection and the existing protection method.

transition resistance	protection method	calculation result/km	protection action result
0.1ohm	Reference (Li et al., 2023a)	-40.5	inaction
	the proposed protection	18.82	action
150ohm	Reference (Li et al., 2023a)	-21.8	inaction
	the proposed protection	9.69	action
300ohm	Reference (Li et al., 2023a)	-15.9	inaction
	the proposed protection	7.81	action

distance protection. The protection method proposed in this paper is to judge the consistency of the opposite side voltage. Although the error of fault distance is 20 % when the near-zone fault occurs, the difference of the opposite side voltage is less than the setting value, and the protection can act correctly.

7. Conclusion

This paper proposes a single-ended protection method based on the composite sequence network diagram and the LVRT strategy of the flexible DC converter. The contributions and innovations of this paper are as follows:

- (1) When the wind farm adopts the negative sequence current suppression control strategy and the flexible DC converter adopts the positive sequence low voltage current limiting control strategy, the obtained algorithm is independent of the transition resistance in principle. The proposed protection method has a strong ability to resist transition resistance. When the transition resistance is as high as 300Ω, the calculation results are still accurate.
- (2) The example analysis shows that when the AC tie-line has an internal fault, the voltage difference fluctuates around 4 kV; When an external fault occurs, the voltage difference is much larger than 4 kV. The voltage difference between internal and external faults is clearly distinguished and easy to set.
- (3) When the near-zone fault of the protection device occurs, the criterion in this paper can operate reliably, which solves the problem of dead zone in the near-zone fault of the traditional single-ended protection based on power frequency. In the case of short-circuit fault at the end of the line, the criterion in this paper can also operate correctly and has good selectivity.

The protection scheme proposed in this paper is suitable for the double-ended weak-infeed system with the sending-end flexible DC converter bus connected to a single wind farm. For the multi-wind farm sending system through VSC-HVDC, it is necessary to further consider the overall equivalent problem of the sending-end flexible DC converter and the wind farm connected to the same bus, as well as the application of the cable in the double-ended weak-infeed system, which will also be the focus of the next research work.

In this paper, the backup protection method of AC tie-line for wind farm through flexible DC transmission system is studied, and a double-ended weak-infeed single-ended protection method based on the analysis of the voltage consistency of the opposite side is proposed. However, due to the limitation of time, further work is still needed in the implementation and promotion of the proposed protection methods. The details are as follows:

- (1) Promote the transformation of the protection method to the application level. The simulation verification of the double-ended weak-infeed single-ended protection method based on the analysis of the consistency of the opposite side voltage proposed in this paper is only based on the digital simulation platform. It is still necessary to implement the principle on the actual relay protection device platform and test the protection performance to further verify the effectiveness of the algorithm.
- (2) Improve the possibility of applying the proposed protection method in more scenarios. The backup protection scheme proposed in this paper is based on single-phase ground fault, which is only applicable to the double-ended weak-infeed system with the sending flexible DC converter bus connected to a single wind farm, and the AC contact line adopts the overhead line model. In the future, the proposed protection scheme needs to be further extended to the multi-wind farm through the flexible DC transmission system, and the application of the cable in the double-ended weak-infeed system needs to be considered.

CRedit authorship contribution statement

Botong Li: Writing – review & editing, Funding acquisition, Formal analysis, Data curation. **Xiaolong Chen:** Resources. **Yuejin Dong:** Writing – review & editing, Writing – original draft. **Jianing Li:** Writing – original draft, Visualization, Validation, Resources. **Shuai Wang:** Validation, Software, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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